

Additional Exercises 2

General Series — Absolute & Conditional Convergence

Mirrors: SIT2 | ★ Easy ★★ Medium ★★★ Hard

Exercise 1: Telescoping series — prove the identity

(a) ★ $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$

(b) ★ $\sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{3}{4}$

(c) ★★ $\sum_{n=2}^{\infty} \ln\left(1 - \frac{1}{n^2}\right) = -\ln 2$

(d) ★★★ $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = \frac{1}{4}$ (use partial fractions)

Exercise 2: Alternating series — classify convergence

For each series, determine: (i) Does it converge absolutely? (ii) Does it converge conditionally (Leibniz test)? (iii) Does it diverge?

(a) ★ $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2 + 1}$

(b) ★ $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

(c) ★ $\sum_{n=1}^{\infty} \frac{(-1)^n(n+1)}{n}$

(d) ★★ $\sum_{n=1}^{\infty} (-1)^n \frac{\ln n}{n}$

(e) ★★ $\sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)}$

- (f) ★★ $\sum_{n=1}^{\infty} (-1)^n \left(1 - \cos \frac{1}{n}\right)$
- (g) ★★ $\sum_{n=1}^{\infty} (-1)^n \left(\sqrt{n+1} - \sqrt{n}\right)$
- (h) ★★★ $\sum_{n=1}^{\infty} (-1)^n \left(1 - \left(\frac{n}{n+1}\right)^n\right)^{1/2}$
- (i) ★★★ $\sum_{n=1}^{\infty} \frac{(-1)^n (2n)!}{4^n (n!)^2}$

Exercise 3: General convergence — classify each series

- (a) ★ $\sum_{n=1}^{\infty} \frac{\cos(2\pi n)}{n^2}$
- (b) ★ $\sum_{n=1}^{\infty} \frac{n}{e^n}$
- (c) ★★ $\sum_{n=1}^{\infty} \frac{1}{n \cdot n^a}, \quad a > 0$
- (d) ★★ $\sum_{n=1}^{\infty} \frac{1}{n \sin(1/n)}$
- (e) ★★ $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{n^3} \cdot \frac{1}{n^3}$
- (f) ★★ $\sum_{n=1}^{\infty} \frac{(2n)!}{4^n (n!)^2}$
- (g) ★★★ $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}$
- (h) ★★★ $\sum_{n=2}^{\infty} \frac{(2n-1)!!}{(2n)!!} \cdot 3^{n/3}$

Exercise 4: Theoretical problems

- (a) ★★ Given $\sum a_n$ converges absolutely. Prove $\sum a_n/n^2$ converges absolutely.
- (b) ★★ Given $\sum a_n$ converges. Does $\sum \sin(a_n/n^3)$ necessarily converge absolutely? Prove or give a counterexample.
- (c) ★★★ Prove the Leibniz test: if $\{a_n\}$ is decreasing, $a_n \geq 0$, and $a_n \rightarrow 0$, then $\sum (-1)^n a_n$ converges, and the error of the N -th partial sum is $\leq a_{N+1}$.

- (d) ★★★ How many terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n(n+1)^2}$ are needed to approximate the sum with accuracy 10^{-5} ?

★ *Direct Leibniz or absolute comparison* ★★ *Limit comparison + Leibniz* ★★★ *Subtle asymptotic analysis or proof required.*